

meter as shown in the circuit diagram. The offset of the circuit can be removed by shorting the test leads with a thick copper wire and setting the set-zero potmeter VR1 for zero deflection in the analogue multimeter. The diodes (D1 and D2) in parallel to the analogue meter are for its protection from over-voltage.

Finally, the voltage output of the

op-amp can be summarised using the relationship for the gain of non-inverting op-amp configuration into a simple relation as:

$$V_o = I_{\text{const}} R_x (R_f/R_i + 1)$$

where I_{const} is the selected constant current from any particular range given here as $V/R = 5V/(500\text{ohms}, 1k, 10k \text{ or } 100k)$ amps. R_f and R_i are given

in the diagram. IC CD4066 (IC2), an analogue switch, is used here to exhibit zero meter deflection when no resistance is connected across the test terminals.

For measuring the low resistance of copper wire (R_x), connect it at appropriate terminals. Select the particular range in such a way that resistance value is indicated on the meter directly. ●

SIMPLE ALERT TIMER

■ T.A. BABU

This timer is a useful device for people who often forget to turn off devices in time. Suppose you want to boil water in your electric kettle or cook something on your gas. You switch on the kettle or the gas and at the same time press switch S2 momentarily to enable the

alert circuit. A ten-minute piezobuzzer reminds you to attend to what you were doing.

The circuit is designed using an N-channel enhancement-mode field effect transistor BS170 (T1). It does not draw any power when inactive and less than 15 microamperes while functioning. A car key 12V fob battery would be an ideal choice in terms of simplicity.



Momentarily pushing switch S2 charges capacitor C1 rapidly via resistor R1. When the voltage on capacitor C1 exceeds the threshold voltage at the gate of T1, it starts charging the reservoir capacitor C2. Transistor T1 remains in conducting state as long

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